Weekly Record W7: Data Governance Framework & 7-Risk Management Register

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Company A: Shell **Company B:** KazMunayGas (KMG)

1. Academic & Theoretical Foundation

In digitally transformed energy enterprises, data serves as a critical strategic asset that requires formal governance, quality control, and risk management architecture. Grounded in enterprise data management frameworks, data governance defines authority, control, and accountability over data assets across their lifecycle.

Deploying artificial intelligence algorithms in heavy industry introduces specific failure modes ranging from model drift and data corruption to cybersecurity breaches and algorithmic bias. Establishing a 5-Stage Data Refinement Workflow paired with a comprehensive 7-Risk Register ensures that operational data remains accurate, secure, and legally compliant before influencing high-stakes industrial decisions.

2. Practical Artifact: 5-Stage Data Workflow & 7-Risk Register

Table 7.1: 5-Stage Data Refinement Workflow for Refinery Operations

Stage #	Workflow Stage Name	Technical Description & Processing Actions	Quality Gate / Governance Control
Stage 1	**Data Ingestion**	Collecting raw IoT streams (pressure, temp, vibration) from refinery units	Outlier detection & telemetry ping validation
Stage 2	**Data Cleaning & Validation**	Removing sensor noise, handling missing fields, schema validation	Automated data quality score check (>95% threshold)
Stage 3	**AI Model Inference**	Running ML models for thermal fouling and anomaly detection	Model confidence threshold check (>85% certainty)
Stage 4	**Human Expert Review**	Refinery engineer evaluates model alerts against operational context	Mandatory engineer sign-off for “Yellow/Red” alerts
Stage 5	**Operational Execution**	Implementing scheduled maintenance or parameter adjustment	Logged execution in SAP Plant Maintenance module

Table 7.2: 7-Risk Register for Industrial AI Deployment at KMG

Risk ID	Risk Category	Specific Risk Description	Likelihood / Impact	Mitigation Strategy & Governance Control
R1	**Data Privacy**	Unauthorized access to employee or contractor operational records	Low / Medium	Strict role-based access control (RBAC) & data anonymization
R2	**Cybersecurity**	External breach of refinery OT network via IoT gateway	Low / High	Air-gapped OT networks & hardware diode protection
R3	**Algorithmic Bias**	AI model trained on historical data misinterprets new crude blends	Med / Medium	Continuous retraining on diverse crude oil assay datasets
R4	**Model Drift**	Sensor degradation causes accuracy loss over time	High / Medium	Quarterly recalibration & model drift auditing protocols
R5	**Accountability Gap**	Unclear legal liability for equipment failure following AI alert	Low / High	Explicit SOP: Named Chief Engineer retains sole legal liability
R6	**Escalation Failure**	System fails to escalate critical anomaly to control room	Low / High	Dual redundant alarm triggers (analog + digital overrides)
R7	**Manual Fallback Risk**	Operators lose manual control skills due to over-reliance on AI	Med / High	Mandatory monthly manual control room drill sessions

3. Comparative Analysis: Shell vs. KazMunayGas (KMG)

1. **Data Architecture & Localization**: Shell utilizes centralized global cloud architectures (AWS/Azure) to aggregate telemetry across international subsidiaries. KMG operates under strict Kazakh data sovereignty legislation, requiring subsoil and operational data to be stored and processed on domestic servers within the Republic of Kazakhstan.
2. **Cybersecurity Environment**: Shell manages a global threat surface requiring complex multi-jurisdictional compliance. KMG's refining data infrastructure relies on isolated, air-gapped operational technology (OT) networks at ANPZ, PNPZ, and PKOP to prevent external cyber threats to national critical infrastructure.
3. **Data Quality Maturity**: Shell maintains mature enterprise data governance with automated data stewards across divisions. KMG is currently building unified data

governance through the ABAI suite, transitioning legacy paper and Excel records into standardized digital formats.

4. Strategic Transferability Verdict

- **ADAPT**: Adapt Shell's comprehensive data governance taxonomy, data stewardship roles, and model performance auditing protocols.
- **REJECT**: Explicitly reject cloud-based offshore data storage for subsoil geological and refinery operational data; enforce 100% domestic data localization compliance under Kazakh law.

5. AI Verification & Level 2 Declaration Log

- **Prompt Used (Gemini 2.5 Pro)**: *"Propose a cloud-based data storage architecture for KMG upstream subsoil data using public international cloud providers."*
- **AI Hallucination Identified**: The AI recommended storing raw seismic data and subsoil logs in public multi-tenant cloud servers located in Western Europe to minimize storage costs.
- **Human Verification & Correction**: **REJECTED AND CORRECTED**. Under Kazakhstan's Law on Subsoil and Subsoil Use and data localization statutes, strategic geological and subsoil data must reside within national boundaries. The architecture was revised to a secure, private domestic cloud infrastructure.

6. Actionable Operational Recommendations & Key Metrics

1. **Data Quality Index**: Maintain a minimum Data Quality Score of 98% across all incoming IoT telemetry at ANPZ.
2. **Audit Frequency**: Institute mandatory quarterly audits of model drift for all active predictive maintenance algorithms.
3. **Fallback Preparedness**: Conduct bi-annual emergency drills testing manual operator takeover of refinery control systems.

7. Reflection, Peer Review & Capstone Linkage

Establishing rigorous data governance and risk management provides the core technical foundation for Attestation A3 (Photo Essay & Risk Framework) and directly informs Capstone (Data Governance & Cybersecurity Controls).